

Windows Server 2022 Active Directory Setup in VirtualBox

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Revision History

Version	Date	Description	Author
1.0	2025-03-13	Initial formatting and procedural enhancements.	IT Operations
1.1	2025-03-17	Updated to reflect typical naming conventions	

Approvals

Name	Role	Signature / Date

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1. Purpose

This procedure outlines the steps for installing Windows Server 2022 inside VirtualBox and configuring it as a Domain Controller with Active Directory Domain Services for lab or testing environments.

2. Scope

This document applies to IT students and staff responsible for setting up test environments using Windows Server 2022 within VirtualBox.

3. Responsibilities

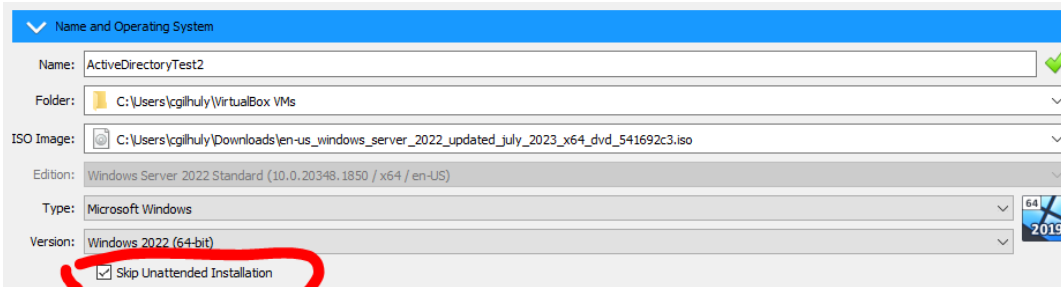
It is the responsibility of the student or assigned personnel to follow this procedure and ensure the server is set up securely and correctly.

4. Prerequisites

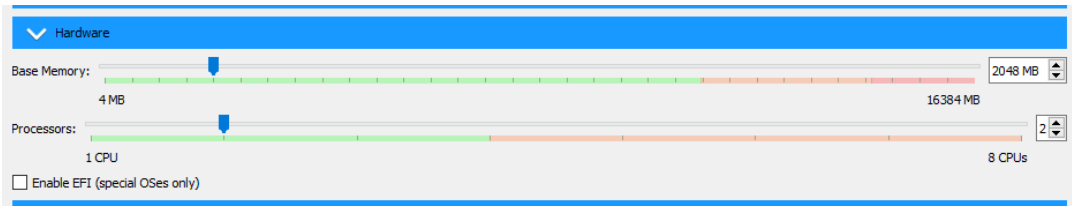
- VirtualBox installed
- Windows Server 2022 Standard ISO (Available in [Azure Education](#))
- Local Administrative privileges
- Basic understanding of virtualization and networking

5. Procedure

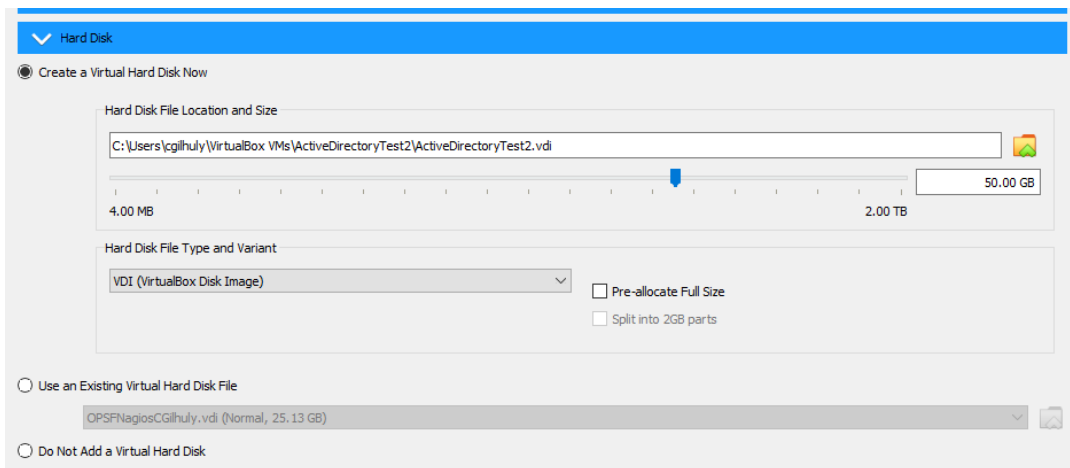
1. Using VirtualBox select the option to add a new Virtual Machine.
2. Include your Name in the VM name as well as other appropriate components. See Appendix for naming convention. For ISO Image choose the location for your Windows Server 2022 Standard ISO file > Make sure you CHECK "Skip Unattended Installation". If you leave this unchecked VirtualBox will install the non-GUI version of Windows Server by default, and you will need to install everything else using PowerShell.....



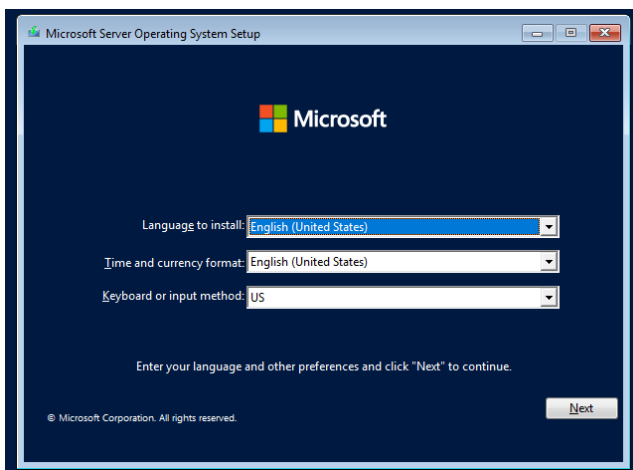
3. Select 2GB of RAM and 2 CPUs should be more than enough for the VM.



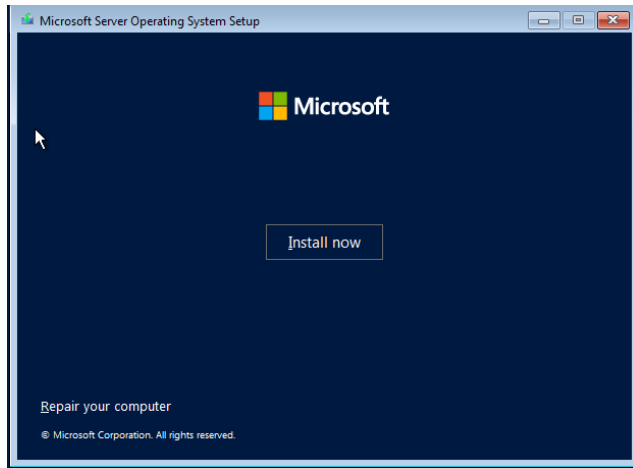
4. The default settings are fine for Hard Disk.



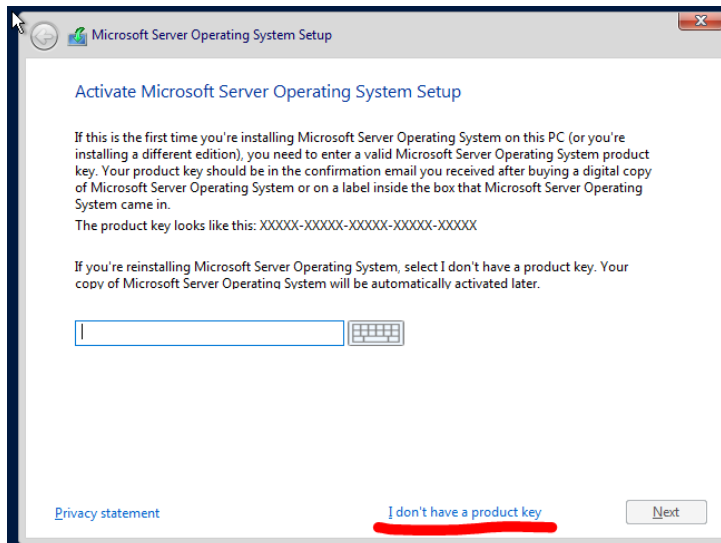
5. Select Finish to add your new VM to VirtualBox > Select your new VM and Start it > Default options are good at the initial MicrosoftServer Operating System Setup screen, click Next



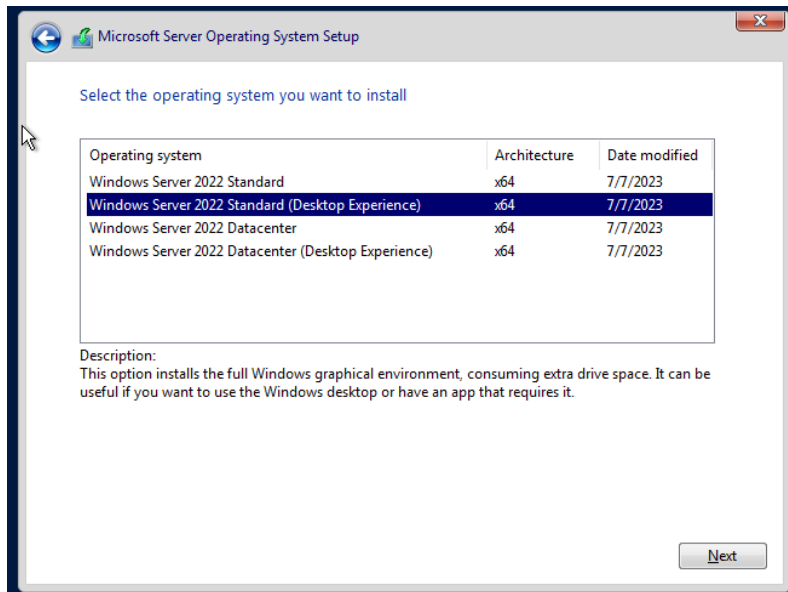
6. Click Install Now



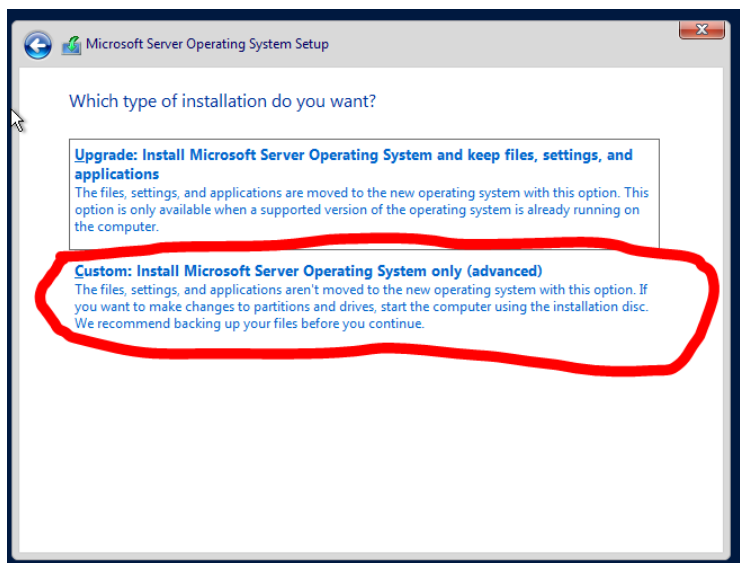
7. For our testing purposes we don't need a product key (although you could get one from your Azure Education Portal if you wanted), click "I don't have a product key"



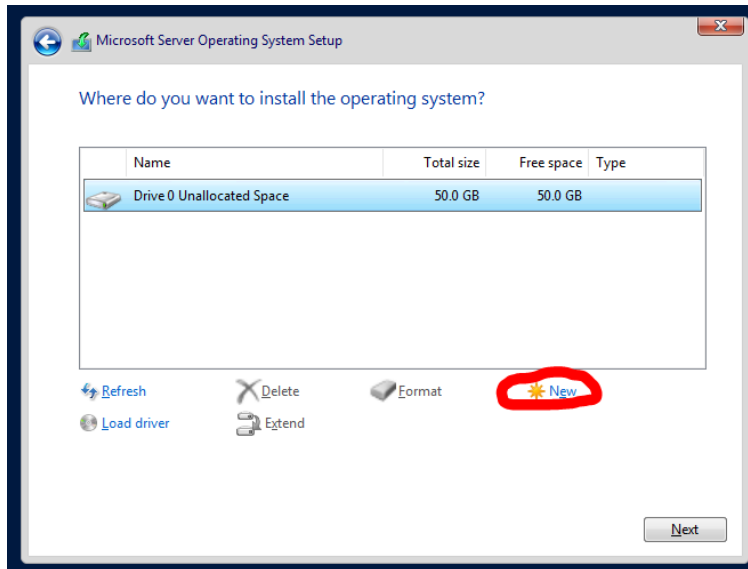
8. **THIS IS VERY IMPORTANT.** Make sure you select the Desktop Experience version of Windows Server 2022 Standard. This is Microsoft speak for GUI version. You want the GUI version.



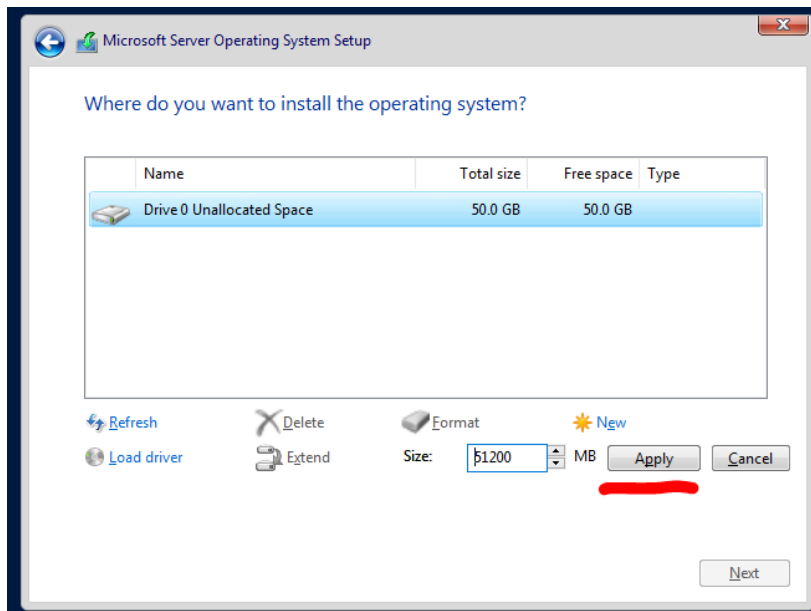
9. Select "Custom: Install Microsoft Server Operating System only".



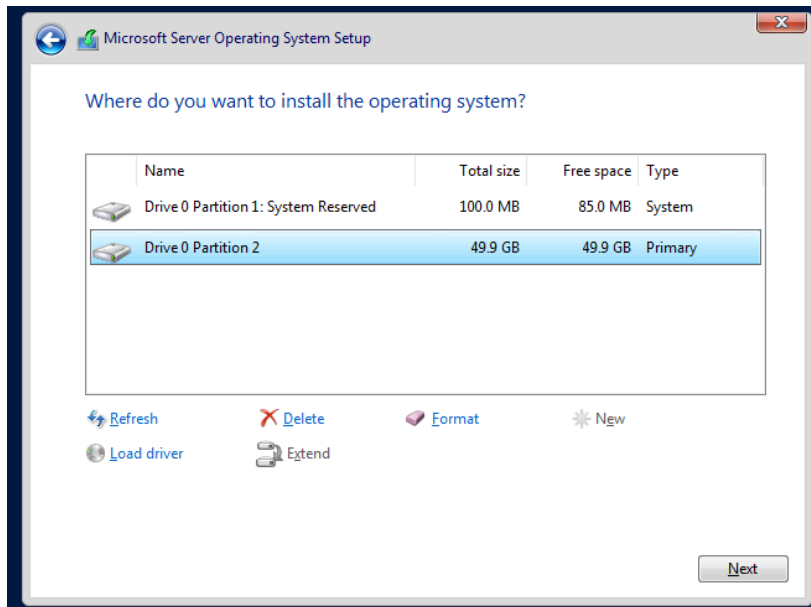
10. Select "New" to create a new partition and allow Windows to automatically create itself a System Reserved partition.



11. Leave the default storage amount and click “Apply” > There will be another box pop up, click “Ok” (or whatever the affirmative option is here)



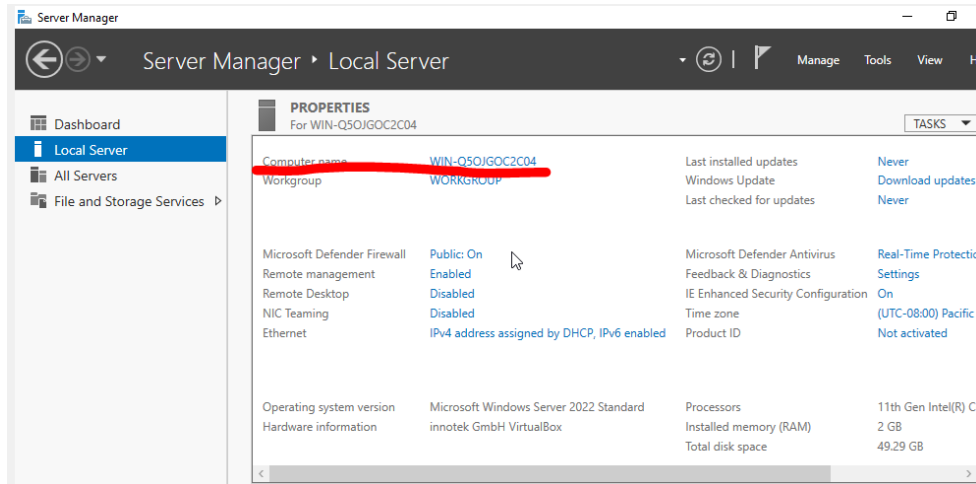
12. Once your new partition is visible click “Next”.



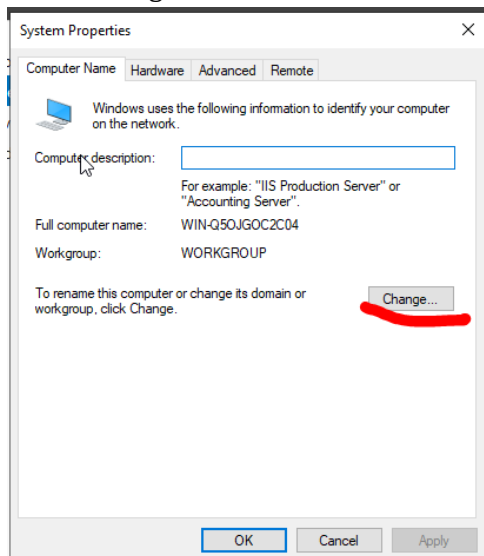
13. Windows will now run through the installation process and then reboot your machine. When it starts back up you will be prompted to create an Administrator password. For our use case make sure the password is something you will remember, as you will need this password multiple times and there will not be a way to reset it.

The screenshot shows the 'Customize settings' window. The title is 'Customize settings'. Below the title is a instruction: 'Type a password for the built-in administrator account that you can use to sign in to this computer.' There are three input fields: 'User name' with the text 'Administrator', 'Password' (empty), and 'Reenter password' (empty).

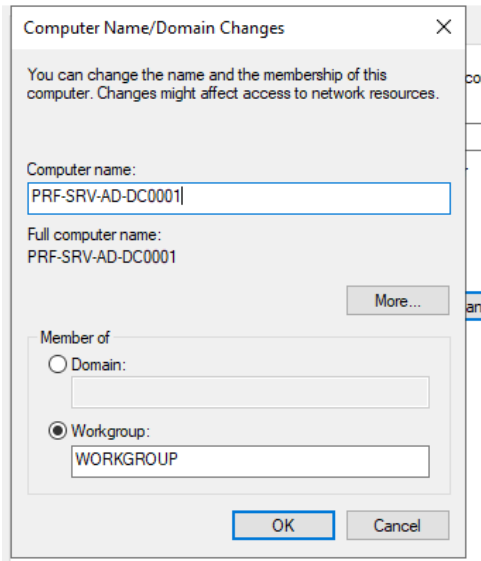
14. Once your server is up and logged in to the Administrator account it will automatically open the Server Manager window. The first thing we are going to do is give our server a proper name. Click "Local Server" > Click on the current "Computer Name"



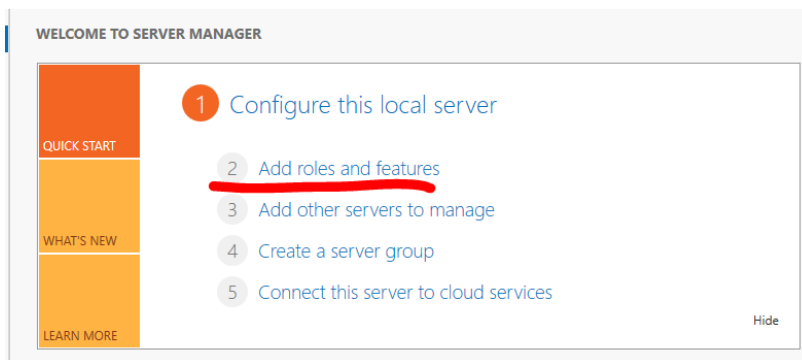
15. Click “Change”



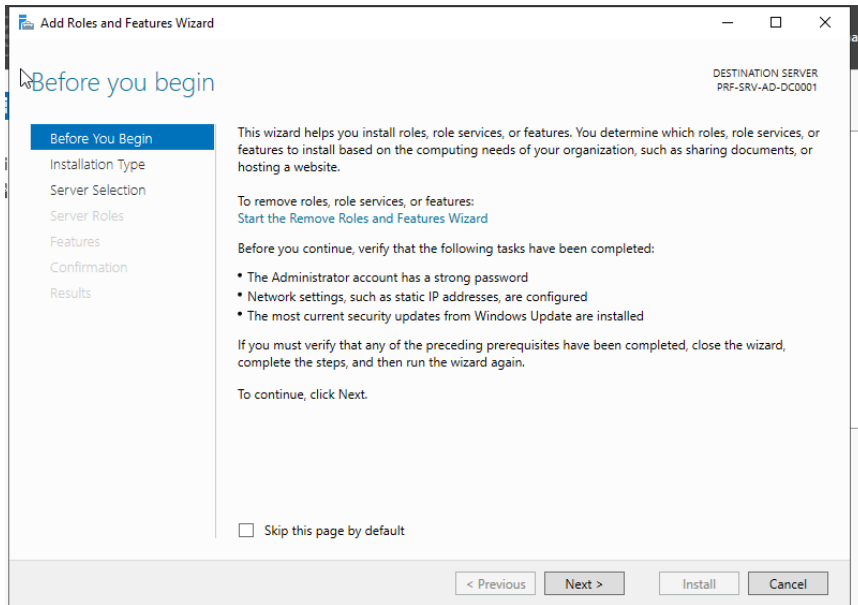
16. Give your server a name that follows proper naming conventions, below you can see the example that I’ve used for mine. > Click “Ok” > A box will pop up that will let you know making this change requires a server reboot, click “Ok” > Another box will pop up after it thinks for a bit asking you to reboot, click “Restart Now”.



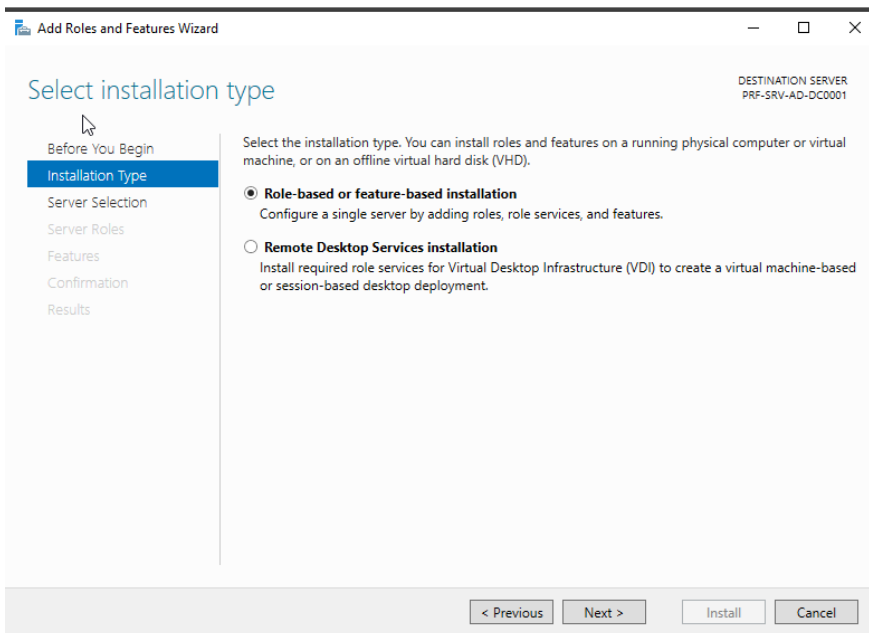
17. Once your server is back up and running you will once again be greeted with the Server Manager window. Click “Add roles and features”



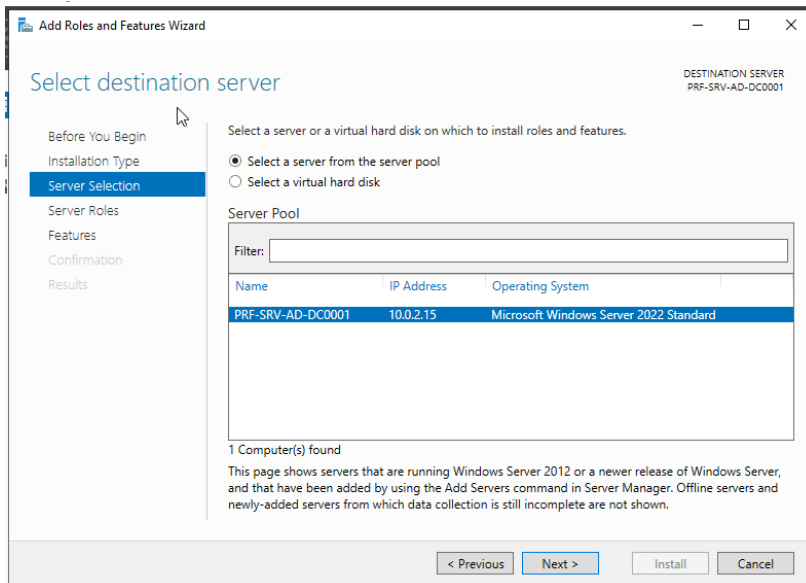
18. Click “Next”



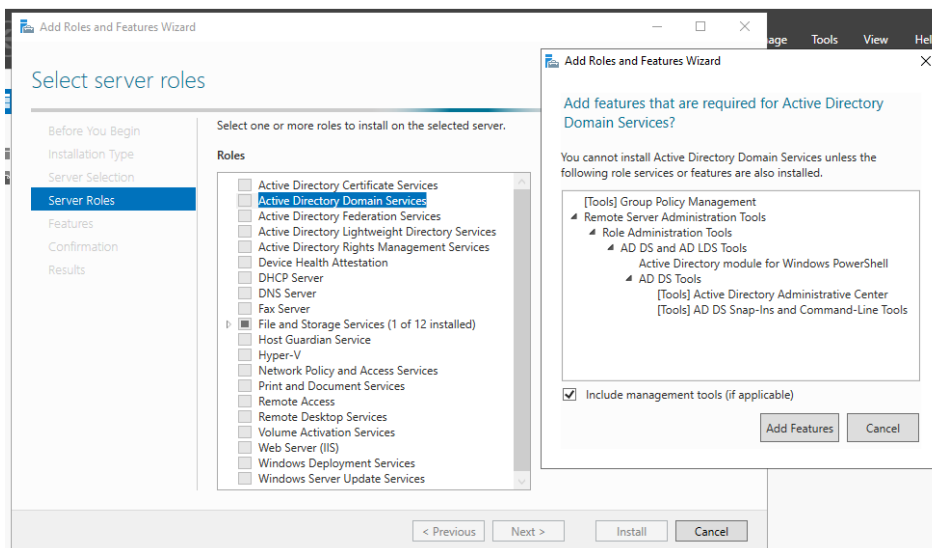
19. Make sure “Role-based or feature-based installation” is selected and click “Next”



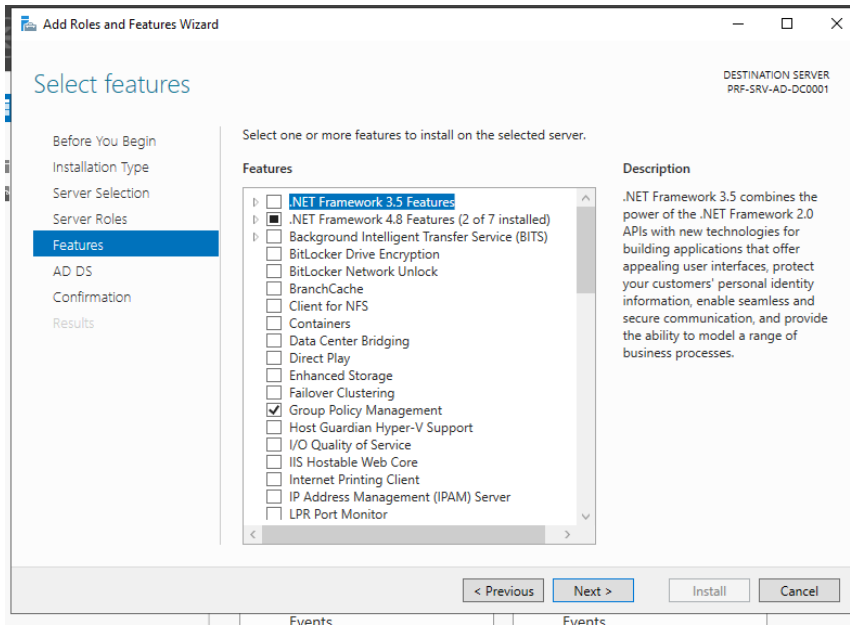
20. There is only one server available, leave that selected and click “Next”.



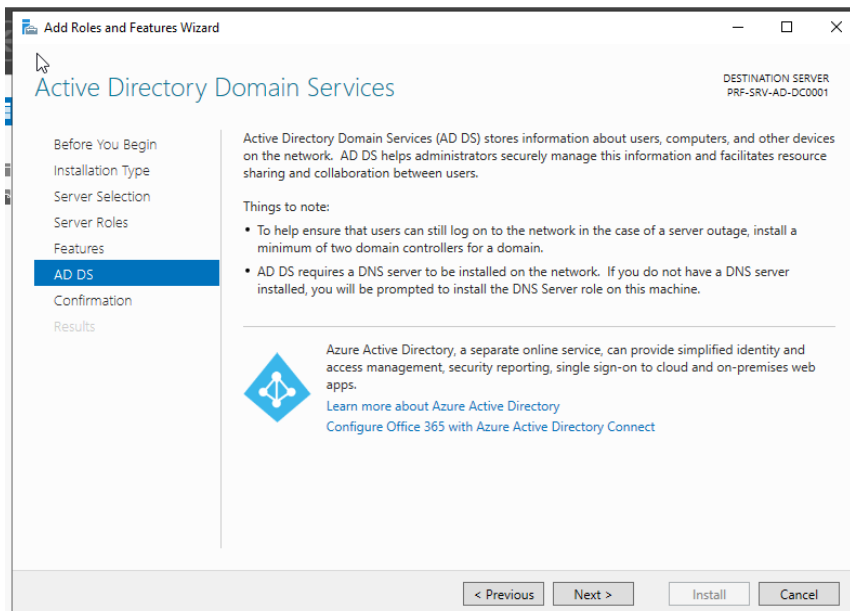
21. Select the “Active Directory Domain Services” box, it will pop up a second window, click “Add Features”



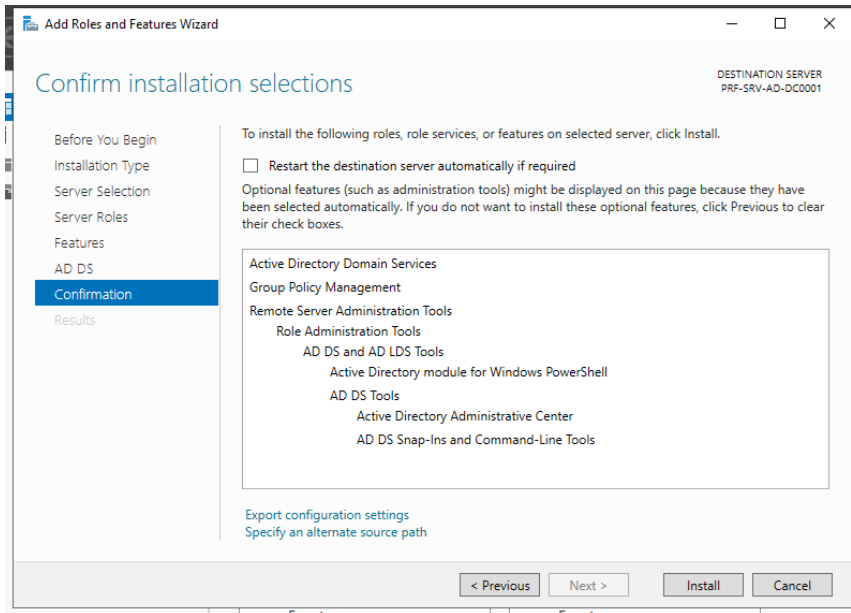
22. Leave the default selections and click “Next”.



23. Click “Next”



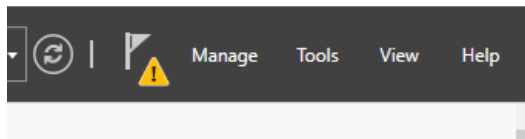
24. Click “Install” to install our new Active Directory Role and Features.



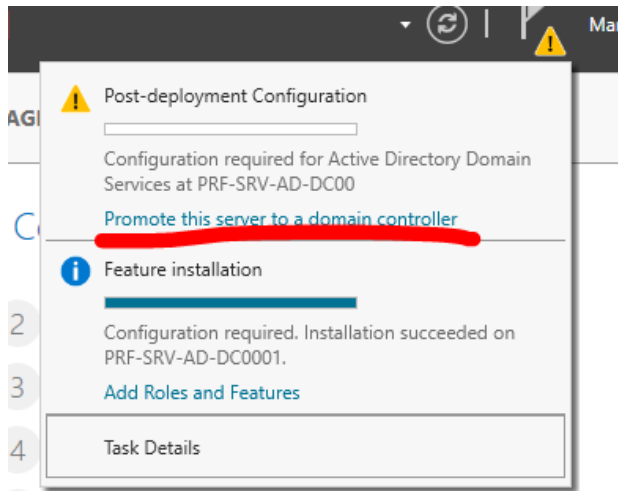
25. Once the progress bar at the top shows that your new Role/Features have been installed you can Close the progress window.

Back on the Server Manager screen you will see a little yellow alert triangle by the Flag symbol.

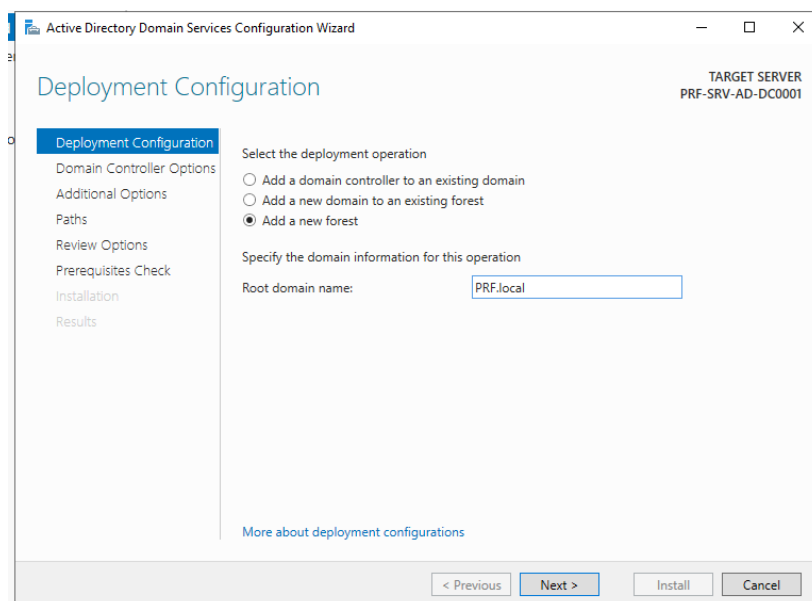
Click on the Flag to drop down a new context box.



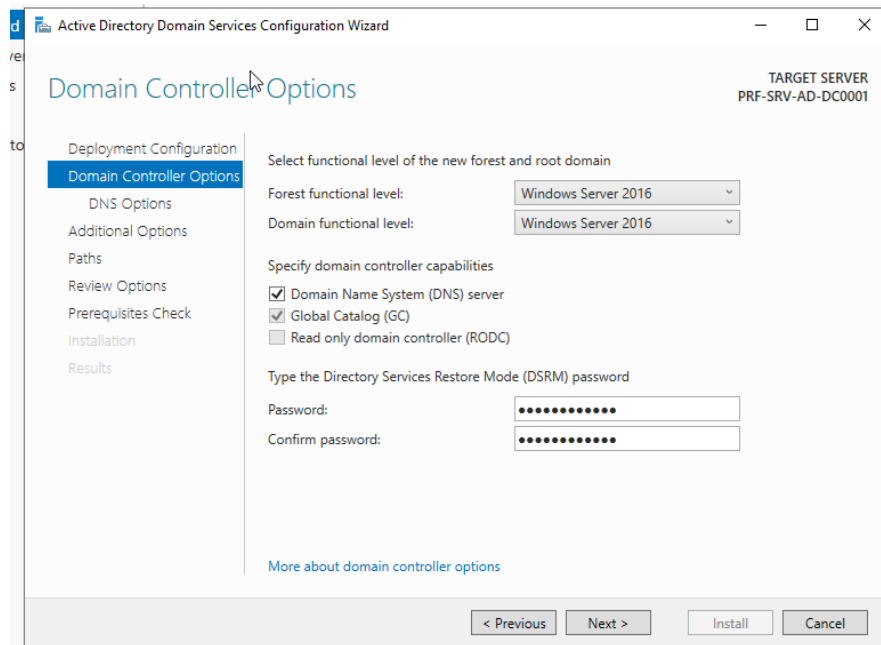
26. Click “Promote this server to a domain controller”



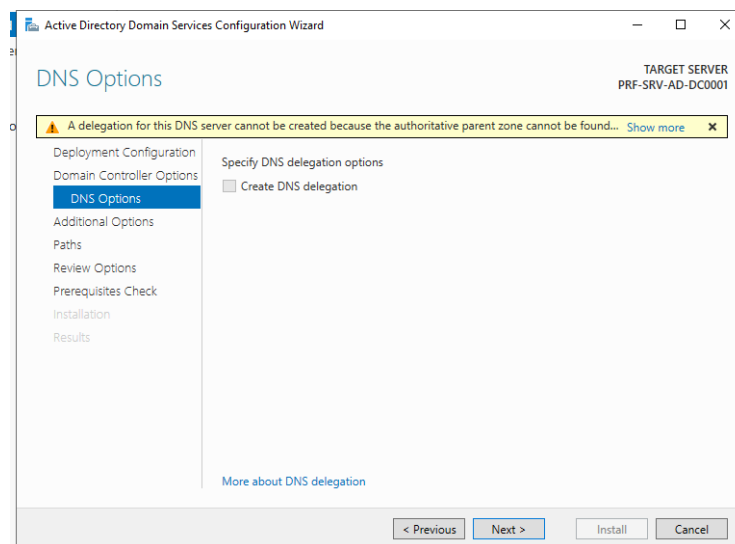
27. Make sure you select “Add a new forest” > Give your root domain a name, it needs to be something followed by “.local” > Click “Next”



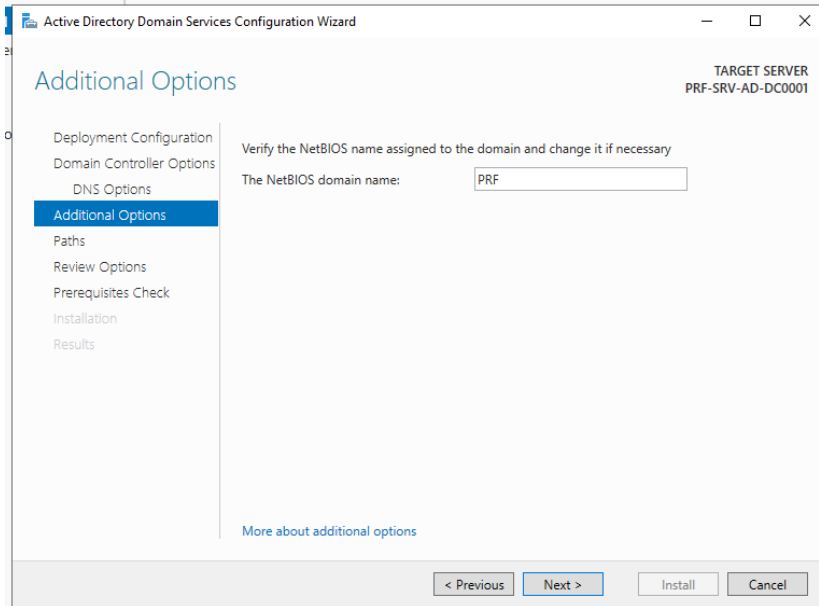
28. Make sure “Domain Name System” is checked > Create a new DSRM password, again make sure it’s something you can remember > Click “Next”.



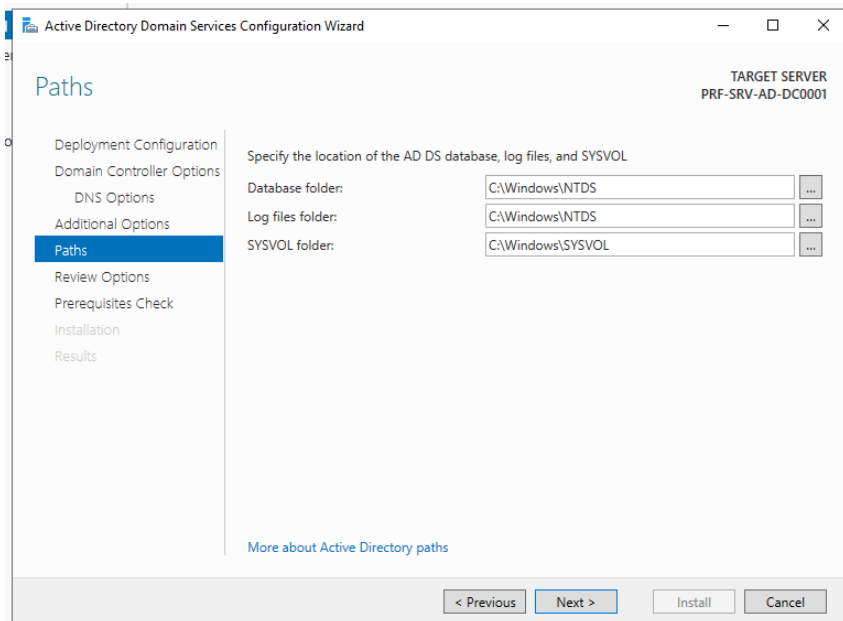
29. Click “Next”



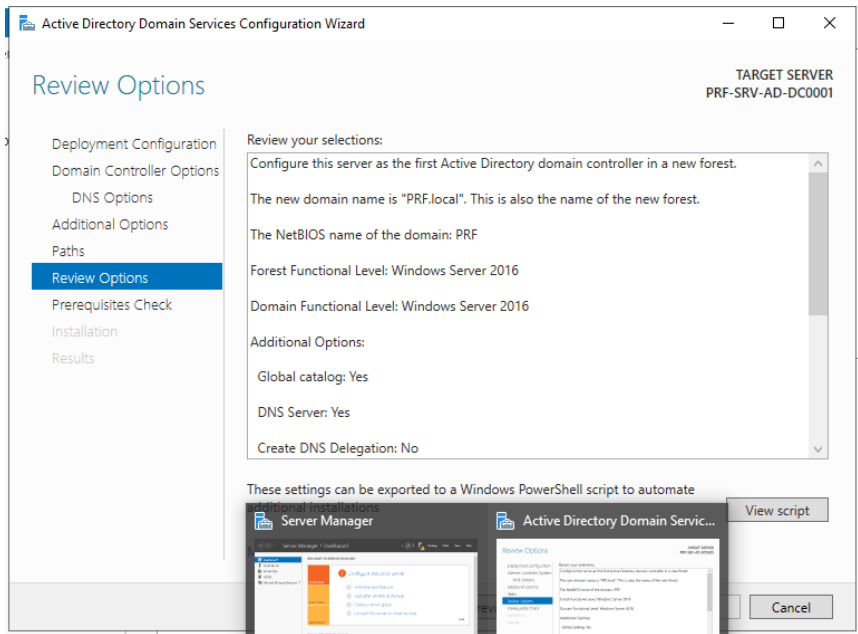
30. The setup should automatically generate your NetBIOS name based on your root domain name, click “Next”.



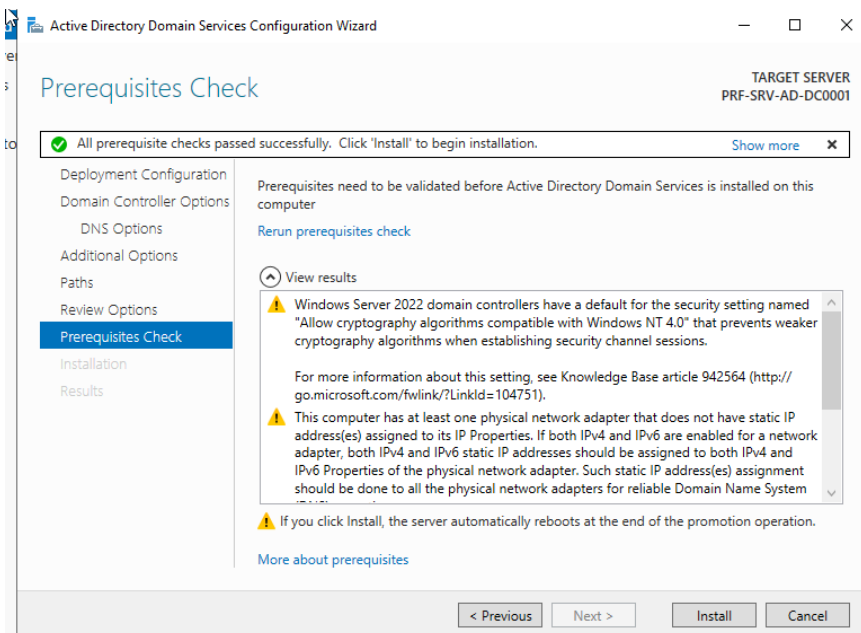
31. Leave the defaults here and click “Next”



32. Review your options and click “Next”.



33. You should get a green check mark and a message that says your Prerequisite Checks have passed successfully. At this time you can safely ignore all those yellow warning notifications, click “Install”



34. Your server will eventually pop up a warning message that says you are going to be logged out, and then it will reboot. Once the server has booted up again you will be able to log back in and access your new Active Directory software.

Congratulations, you've created a Windows 2022 Server and successfully installed Active Directory!

6. Additional Notes

Take a Snapshot (Recommended)

Once you've successfully completed the Active Directory setup, take a VirtualBox snapshot. This will allow you to return to this working state if anything breaks later.

Configure Network Adapter (Optional)

Before starting your VM, it's important to configure the network adapter to ensure the VM can communicate with your host or other VMs. In VirtualBox, go to the VM settings > Network > Adapter 1. Choose 'Bridged Adapter' or 'Host-Only Adapter' depending on your lab setup.

Logging In After AD Setup

After promoting your server to a domain controller and rebooting, log in using the domain format. Your username will now be 'YOURDOMAIN\Administrator'. Replace 'YOURDOMAIN' with the domain name you created earlier (e.g., svrname.local\Administrator').

Check System Time

Ensure your system time is accurate by clicking the clock in the taskbar. Active Directory depends on time synchronization, especially when working in environments with multiple machines.

7. Server Naming Convention

A solid server naming convention can make managing infrastructure way easier, especially as your environment grows. A good convention should be:

- Descriptive – You should be able to tell what the server is for at a glance.
- Consistent – Stick to the format across environments.
- Scalable – Works for dozens or thousands of servers.
- Environment-aware – Helps you avoid messing up prod when you're in dev.

Common Elements to Include:

[Company/Org]-[Env]-[Location]-[Role/Function]-[Instance/Number]

Examples:

Segment	Example	Meaning
Company	PRF	Pixell-River Financial
Environment	PRD / DEV / TST / UAT / STG	Production, Development, etc.
Location	TOR / MTL / YWR / CAN	Data center or cloud region
Role / Purpose	DB / APP / WEB / FS / DC	Database, App Server, Web, File Server, Domain Controller
Instance #	01, 02, A, B, etc.	Uniqueness & scaling

Sample Names

PRF-PRD-TOR-DB-001 → Pixell-River production database server in Toronto.

PRF-DEV-MTL-WEB-002 → Development web server #2 in Montreal.

PRF-STG-CAN-APP-A → Staging App server in a Canadian cloud region.

PRF-PRD-TOR-DC-01 → Domain Controller for production in Toronto.

Tips

- Use hyphens (-) or underscores (_) only — no spaces.

- Keep names short but informative (ideally ≤ 30 characters if systems like Active Directory are involved).
- Avoid embedding sensitive info (e.g., IP addresses or credentials).
- If using cloud infrastructure, consider including provider codes like AWS, AZR, or GCP.

8. Standard Environments

Each environment in a server infrastructure plays a specific role in the software development lifecycle (SDLC). Here's a breakdown of common environments and what they are typically used for.

DEV (Development)

- Where developers build and test new features or code.
- Often changes rapidly and may contain unstable or experimental code.
- Usually has mock or test data, not real customer information.
- May be set up on local machines or shared cloud/dev servers.

TST (Testing)

- Used for formal testing after development work is complete.
- Typically includes automated tests, QA validation, and bug fixing.
- More stable than DEV but still may be reset or rebuilt frequently.
- Can include both unit and integration testing environments.

UAT (User Acceptance Testing)

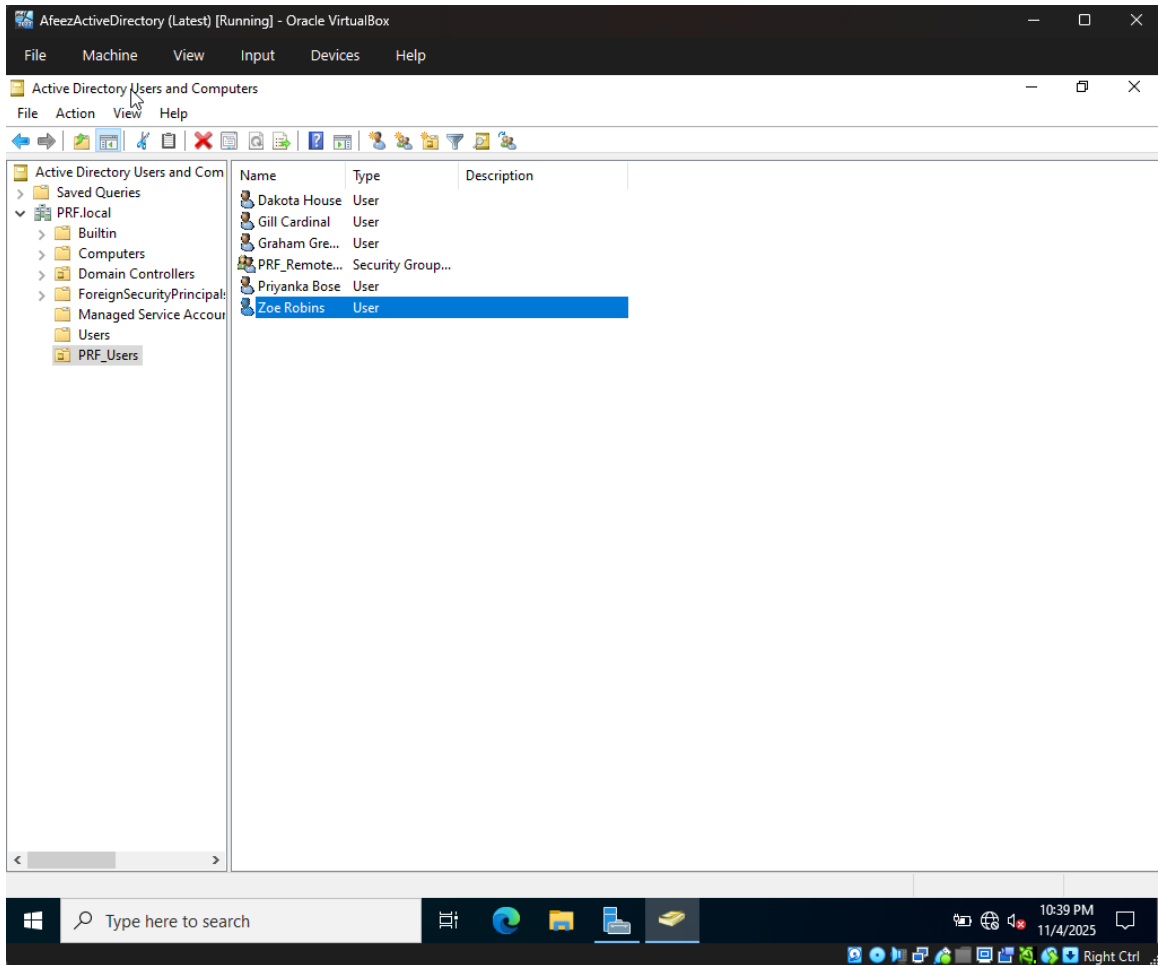
- Used by end users or business stakeholders to validate functionality.
- Tests whether the software meets business requirements and expectations.
- Typically mirrors staging or production closely in terms of data and setup.
- Final checkpoint for sign-off before production deployment.

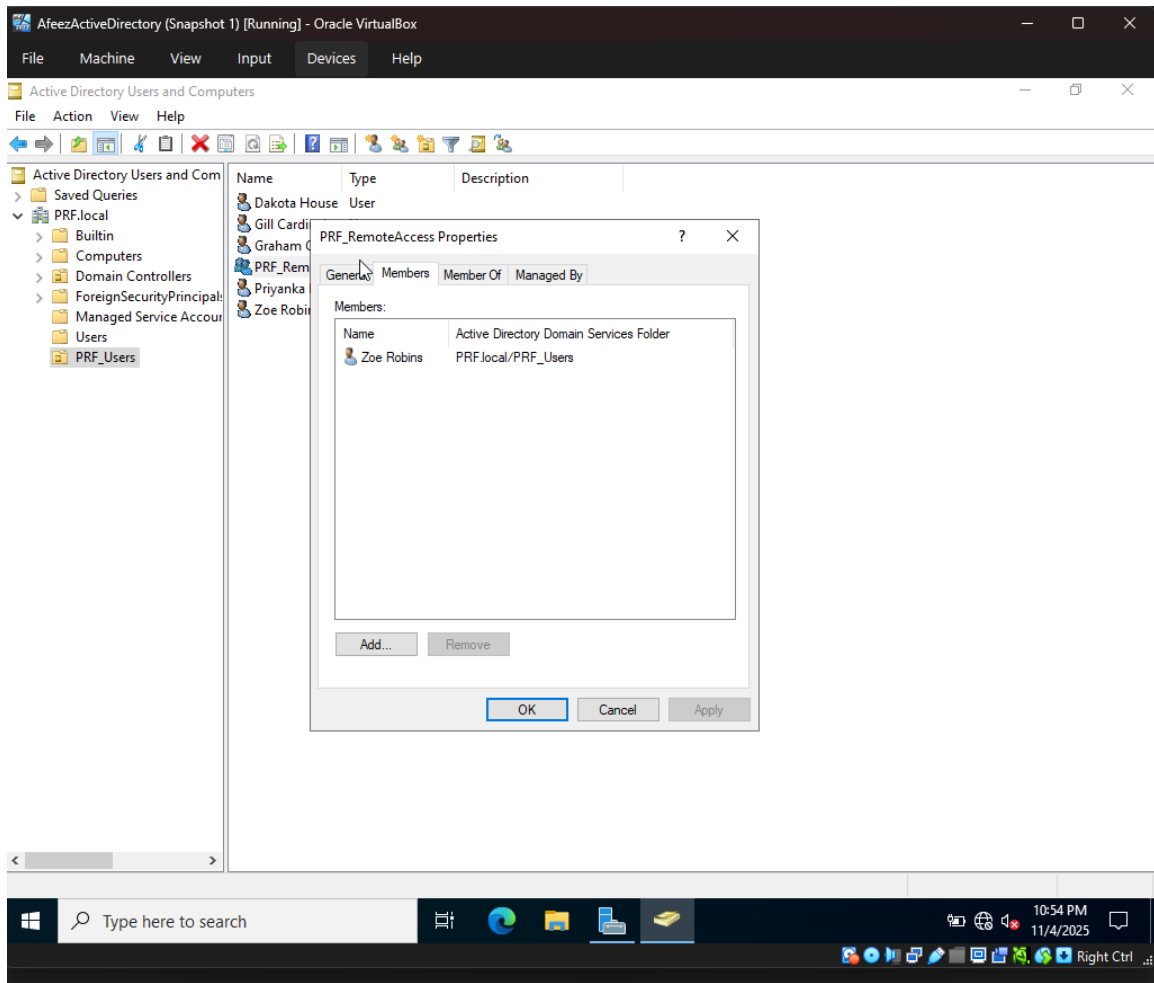
STG (Staging)

- Pre-production environment that closely mirrors the production setup.
- Used for final testing, integration, and deployment rehearsal.
- Often used for performance testing, UAT, and catching last-minute issues.
- Should simulate real-world usage as closely as possible.

PRD (Production)

- Live environment where the application is accessed by end users.
- Highly stable, secure, and monitored closely.
- Changes are tightly controlled through formal deployment processes.
- Real customer data and mission-critical workloads are handled here.





Full Name	Username	User Principal Name (UPN)	Temporary Password
Zoë Robins	zrobins	zrobins@prf.local	Pr2025!Zr#1
Priyanka Bose	pbose	pbose@prf.local	Pr2025!Pb#2
Gil Cardinal	gcardinal	gcardinal@prf.local	Pr2025!Gc#3
Dakota House	dhouse	dhouse@prf.local	Pr2025!Dh#4
Graham Greene	ggreene	ggreene@prf.local	Pr2025!Gg#5